UTM
UNIVERSITI TEKNOLOGI MALAYSIA

# UNIVERSITI TEKNOLOGI MALAYSIA

## BACHELOR in SOFTWARE ENGINEERING (SCSEH)





UTM JOHOR BAHRU

*Innovating Solutions*

# AGENDA

01 PROGRAM ADMINISTRATION

02 PROGRAMME INFORMATION

03 THINGS A STUDENT MUST KNOW



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# 01 PROGRAM ADMINISTRATION

# UG ACADEMIC ADMINISTRATION BEARERS



**Prof. Dr. Dayang Norhayati**
**Academic & Student Affairs**



**Mr. Masnawi Bin Miskam**
**Deputy Registrar**



**Prof. Ts. Dr. Md Asri bin Ngadi**
**Computer Science**

**SECRH**



**Dr. Sharin Hazlin binti Huspi**
**Applied Computing & Artificial Intelligence**

**SECPH**
**SECBH**



**AP Dr. Radziah Mohamad**
**Software Engineering**

**SECJH**
**SCSEH**



**AP Ts. Dr. Farhan bin Mohamed**
**Emergent Computing**

**SECVH**

# COORDINATORS



**Assoc. Prof. Dr. Radziah Mohamad**
**Director (Software Engineering)**
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**Ts. Dr. Johanna Ahmad**
**Coordinator**
**Bachelor in Software Engineering**
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**Dr. Shahliza Abdul Halim**
**Coordinator**
**Bachelor of Computer Science (Software Engineering)**
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# UG ACADEMIC OFFICE STAFF



MISS NOREHA BINTI MAT ISA @ ABDUL RAHMAN



MR. ABDULLAH YUSOF



MS. NOORINA BTE MD. ZEIN



MS. AZMANIZA AZIZ



MS. ROHANI MOHD. ZAIN



MS. NURUL AIDA BINTI JUNAIDI

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# 02 PROGRAMME INFORMATION

# Program Structure

- **Name of award:**
  - **SCSEH: Bachelor in Software Engineering with Honors**

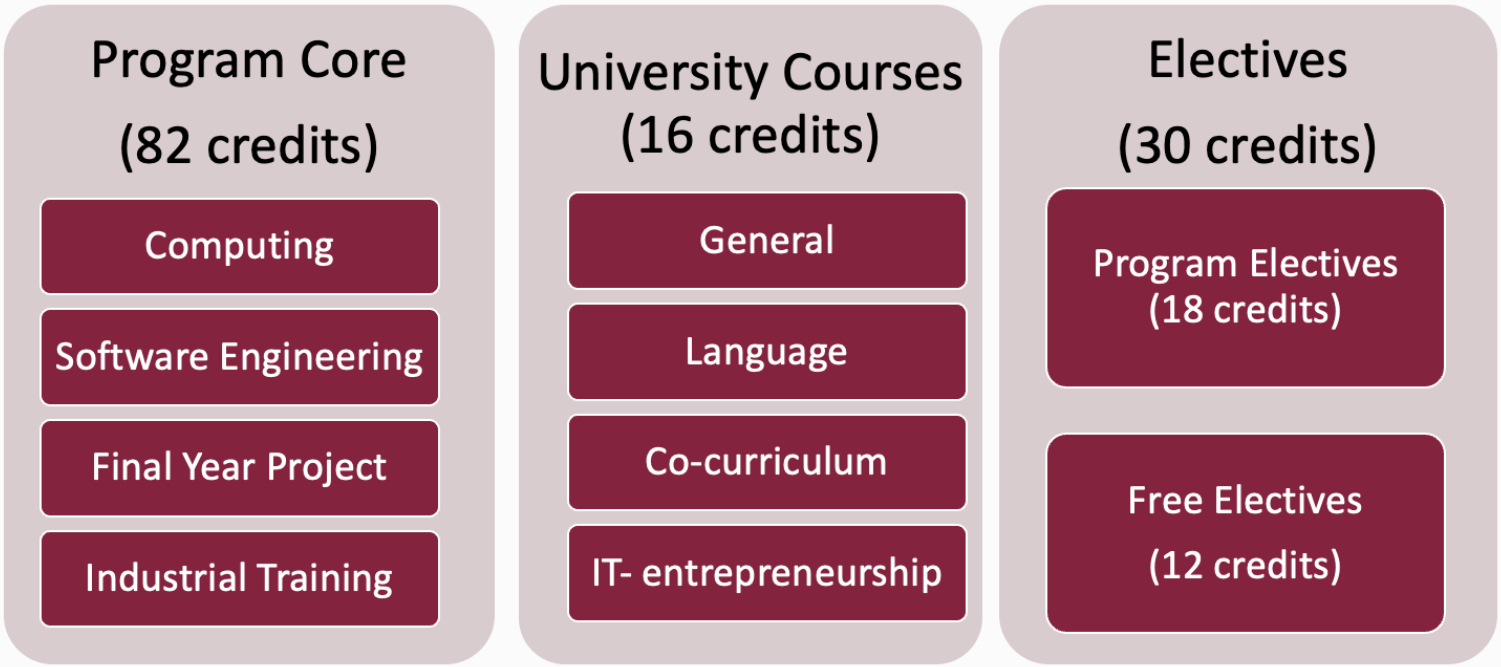
- **Study duration:**
  - **Minimum : 4 years (8 semesters)**
  - **Maximum: 6 years (12 Semesters)**

# Program Structure (cont.)

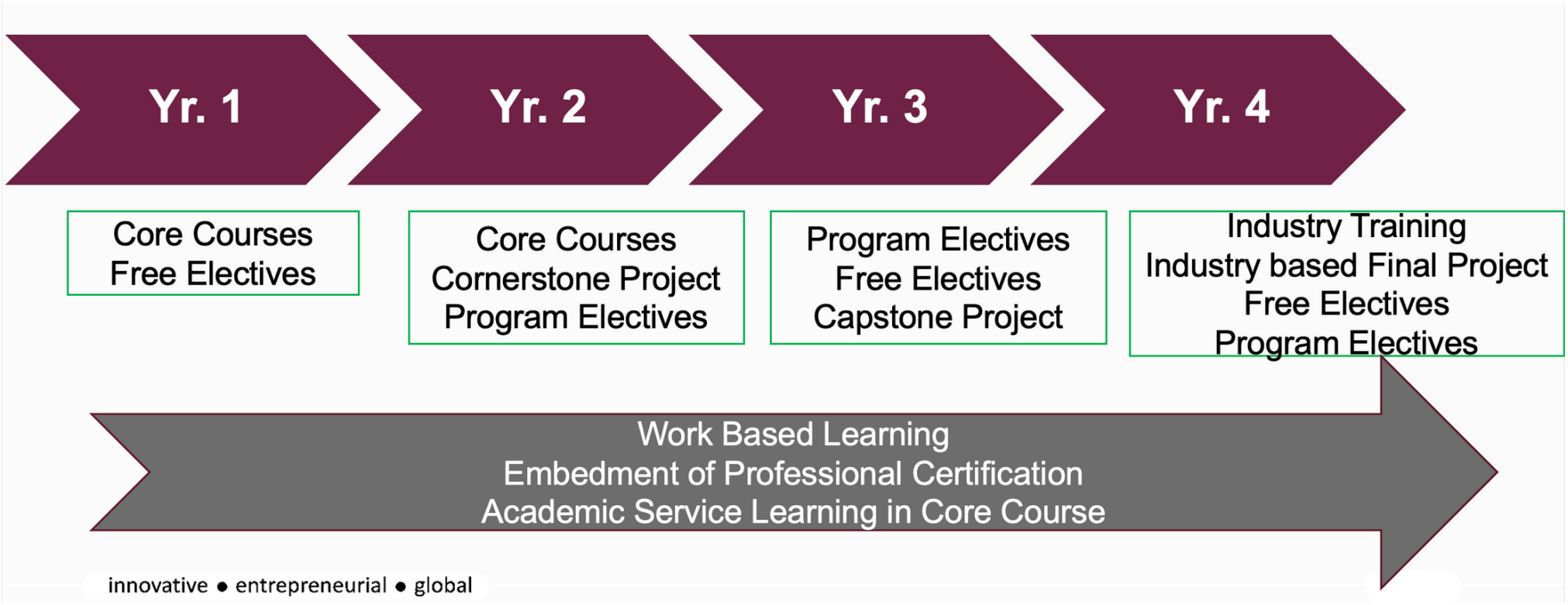
Award Requirements

- To graduate, students must:
  a. Achieve a total of **128 credit hours** with **minimum CGPA of 2.0**.
  b. Pass industrial training (equivalent to 12 credit hours), which 4 credits will be graded and 8 credits as HW status.
  c. Complete Software Engineering Project.
  d. Pass 5 Professional Skills Certificate (PSC).

# Bachelor in Software Engineering (BSE) COMPONENTS



# BSE Industry-Infused Components



# Curriculum Structure Bachelor in Software Engineering

* not including University Core Courses (16 credits)

***# courses in blue are representing WBL courses***





| Year 1 | Year 2 | Year 3 | Year 4 |
|---|---|---|---|
| ◦Fundamental Programming Concepts<br>◦Digital Logic<br>◦Mathematics for Software Engineer<br>◦Database Engineering<br>◦Computer Organisation and Architecture<br>◦Theoretical Foundation of Software Engineering<br>◦Advanced Programming<br>◦Network Communications<br>◦**Software Engineering Principles**<br>◦**Free Elective Course #1** | ◦Operating Systems<br>◦Data Structure and Algorithm<br>◦**Human Computer Interaction Fundamental**<br>◦**Software Process and Project Management**<br>◦**Cross-Platform Application Development**<br>◦Probability and Statistical Data Analysis<br>◦Software Design and Architecture<br>◦Application Development Project I<br>◦Elective Course #1 | ◦Secure Software Programming<br>◦Application Development Project II (Capstone)<br>◦Software Quality & Testing<br>◦**Professional Practice in Software Engineering**<br>◦Free Electives #2<br>◦Elective Course #2<br>◦Elective Course #3<br>◦Elective Course #4<br>◦Software Engineering Project I | ◦**Industrial Training**<br>◦**Industrial Training Report**<br>◦**Software Engineering Project II**<br>◦**Free Elective Course #3**<br>◦**Free Elective Course #4**<br>◦**Elective Course #5**<br>◦**Elective Course #6** |

# Course Registration

- Students MUST register for all the courses to be taken in that semester.
- Students MUST pre-register their courses for the following semester through online system within the registration period given by the university.
- Must discuss and get approval from the academic advisor.

# ACADEMIC WORKLOAD EVERY SEMESTER

### Students with Good Status (KB)

- Min Credits: 12
- Max Credits: 18
- More than EIGHTEEN (18) credits have to seek approval from the Dean of the Faculty

### Students with Probation Status (KS)

- Min Credits: 9
- Max Credits: 12

### Short Semester

- Max: 8 credits

Students are **NOT** allowed to take **more than TWENTY-ONE (21) credits** in a semester

# Academic Standing

| MARKS | GRADE | POINTS | LEVEL OF ACHIEVEMENT |
|---|---|---|---|
| 90-100 | A+ | 4.00 | EXCELLENT |
| 80-89 | A | 4.00 | |
| 75-79 | A- | 3.67 | |
| 70-74 | B+ | 3.33 | GOOD PASS |
| 65-69 | B | 3.00 | |
| 60-64 | B- | 2.67 | PASS |
| 55-59 | C+ | 2.33 | |
| 50-54 | C | 2.00 | |
| 45-49 | C- | 1.67 | |
| 40-44 | D+ | 1.33 | |
| 35-39 | D | 1.00 | FAIL (UM) |
| 30-34 | D- | 0.67 | |
| 00-29 | E | 0.00 | |

# Academic Standing (cont.)

- A student's academic standing is determined at the end of every regular semester based on CGPA.

| CGPA | Academic Standing |
|---|---|
| CGPA > 2.00 | Good Standing (KB) |
| 1.70 ≤ CGPA < 2.00 | Probationary Standing (KS) |
| CGPA < 1.70 | Failure Standing/Academic Dismissal (KG) |

- Students who obtained **THREE (3) consecutive Probation Status (KS)** will be given a **Fail Status (KG)** and they will be terminated from their studies.

# Academic Standing (cont.)

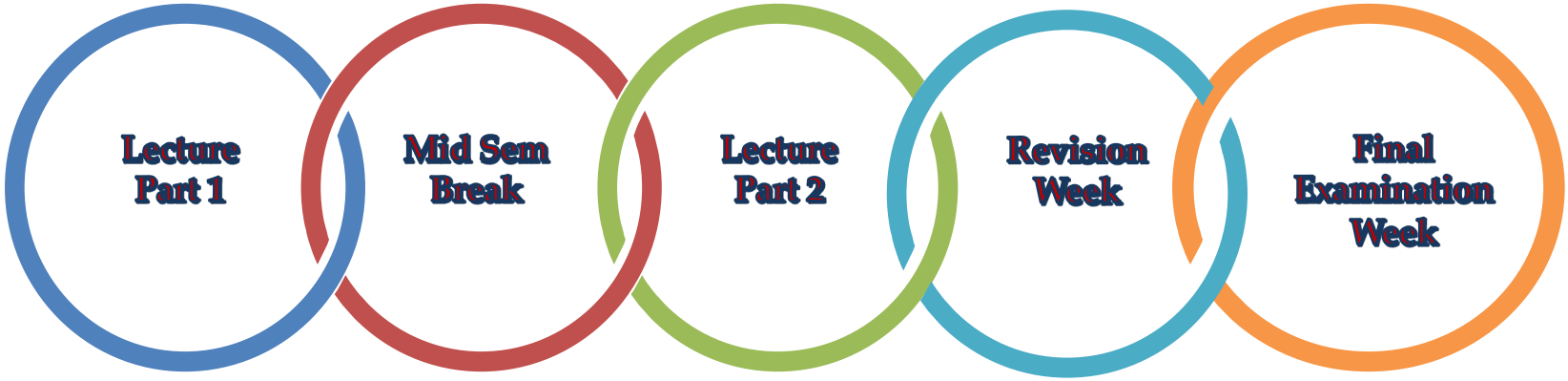
- A student with GPA < 1.00 even though CGPA > 1.70 may continue with their studies. However, the Senate can:
  - i. defer their studies to the following semester; OR
  - ii. give a Fail Status (KG) and terminate their studies.

- Dean List
  - GPA >=3.67 (register 12 credits excluding HS, HW, UG, UK courses) à Dean List certificate.
  - Graduating students (CGPA >=3.67) à Dean List Medal.
  - Appears in transcript

# Replacement Grade

- A student is allowed to improve his academic performance with the permission of the school with the following conditions:
  - a repeat of a passed course with (**grade B- and below**)
  - Only **ONCE** for every course.
  - **Max. credit** allowed for improving grade is **15 credits**
  - A better grade will be counted for CGPA.
  - Only for courses offered in the semester.
  - A fee will be charged for every credit. (Refer to Bursary)
  - Refunds will not be given if the student withdraws from the course.

# Academic Calendar Sem I 2024/25




| Lecture Part 1 | Mid Sem Break | Lecture Part 2 | Revision Week | Final Examination Week |
|---|---|---|---|---|
| 6 Oct – 23 Nov | 24 – 30 Nov | 1 Dec 24 – 18 Jan 25 | 19 – 25 Jan 25 | 31 Jan – 16 Feb 25 |
| 7 weeks | 1 week | 7 weeks | 1 weeks | 3 weeks |

# SCSEH Year 1 Structure

| YEAR 1: SEMESTER 1*** | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE1013 | Fundamental Programming Concept | 3 | |
| SCST1123 | Mathematics for Software Engineer | 3 | |
| SCSR1013 | Digital Logic | 3 | |
| SCST1143 | Database Engineering | 3 | |
| URLS1032 | Integrity and Anti-Corruption | 2 | |
| Sxxxxxx3 | Free Elective I* | 3 | |
| | **TOTAL CREDIT** | **17** | |
| | **CUMULATIVE CREDITS** | **17** | |

****Students with IELTS Band less than 5.5 or TOEFL less than 525 or TOEFL IBT less than 60 or CEFR less than B2 or MUET less than Band 4 {Malaysian student) must register for UHLB1112 course*

** Students must choose University Free Electives subjects offered by faculties other than Faculty of Computing.*

| YEAR 1: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE1203 | Software Engineering Principles | 3 | SCSE 1013 |
| SCSR1033 | Computer Organisation and Architecture | 3 | SCSR1013 |
| SCST1223 | Probability and Statistical Data Analysis | 3 | |
| SCSE1224 | Advanced Programming | 4 | SCSE1013 |
| SCSR2213 | Network Communications | 3 | |
| Uxxxxxx2 | University Elective II | 2 | |
| ***Malaysian Students*** | | | |
| UHMS1182 | Appreciation of Ethics and Civilizations | | |
| ***International Students*** | | | |
| UHLM1012 | Malaysia Language for Communication | | |
| | **TOTAL CREDIT** | 18 | |
| | **CUMULATIVE CREDITS** | 35 | |

# List of Free Electives Sem I 2024/25 - Choose 1

- SEAA1213 Engineering Mechanics
- SSCT1613 Microbiology
- SFMD1063 Information Technology Business
- SSPB2133 International Business
- SKEB1513 Human Anatomy and Physiology
- SBEA1223 Basic Architectural Computing
- SBEQ1343 Introduction to Quantity Surveying
- SBEZ1652  Introduction to Landscape Architecture

# SCSEH Year 2 Structure

| YEAR 2: SEMESTER 1 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE2133 | Software Process and Project Management | 3 | SCSE 1203 |
| SCSE2123 | Software Requirements Engineering | 3 | SCSE 1203 |
| SCSE2103 | Data Structure and Algorithm | 3 | SCSE1013 |
| SCSR2043 | Operating System | 3 | |
| SCSM2113 | Human Computer Interaction | 3 | SCSE 1203 |
| UKQF2xx2 | Service Learning & Community Engagement Courses | 2 | |
| | **TOTAL CREDIT** | **17** | |
| | **CUMULATIVE CREDITS** | **52** | |

| YEAR 2: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSM2213 | Cross-Platform Application Development | 3 | SCST 1143 |
| SCSE2233 | Software Design & Architecture | 3 | SCSE1203 |
| SCSE2243 | Application Development Project I | 3 | SCSE1203, SCSE2123 |
| UHLB2122 | Professional Communication Skills1 | 2 | |
| UHIS1022 | Philosophy and Current Issues | 2 | |
| **Program Electives – Choose 1 (3 credits)** | | | |
| SECB2103 | Bioinformatics I | 3 | |
| SECP2753 | Data Mining | 3 | |
| SECP3213 | Business Intelligence | 3 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **68** | |

# SCSEH Year 3 Structure

| YEAR 3: SEMESTER 1 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| UHLB3132 | Professional Communication Skills 2 | 2 | |
| UHLx1122 | Foreign Language Elective | 2 | |
| Sxxxxxx3 | Free Elective II* | 3 | |
| **Program Electives – Choose 3 (9 credits)** | | | |
| SCST3223 | Data Analytic Programming | 3 | |
| SCSE3143 | Ubiquitous Computing | 3 | |
| SCSR3113 | Cloud Computing | 3 | |
| SCSE3103 | Cognitive Computing | 3 | |
| SCSE3203 | Special Topics | 3 | |
| SCSM3113 | Virtual and Augmented Reality Application | 3 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **84** | |

** Students must choose University Free Electives subjects offered by faculties other than Faculty of Computing.*

| YEAR 3: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE3242 | Software Engineering Project I | 2 | 80 credits<br>SCSE2243 |
| SCSR3133 | Secure Software Programming | 3 | SCSM 2213 |
| SCSE3213 | Software Quality & Testing | 3 | SCSE 2123<br>SCSE 2233 |
| SCSE3223 | Application Development Project II | 3 | SCSE2243 |
| SCSE3233 | Professional Practice in Software Engineering | 3 | SCSE 1203 |
| UBSS1032 | Entrepreneurship and Innovation | 2 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **100** | |

# SCSEH Year 4 Structure

| YEAR 4: SEMESTER 1 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE4108 | Industrial Training (HW) | 8 | 92 credits CGPA >= 2.0 |
| SCSE4114 | Industrial Training Report | 4 | 92 credits CGPA >= 2.0 |
| | **TOTAL CREDIT** | **12** | |
| | **CUMULATIVE CREDITS** | **112** | |

| YEAR 4: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE4214 | Software Engineering Project II | 4 | SCSE3242 |
| Sxxxxxx3 | Free Elective III | 3 | |
| Sxxxxxx3 | Free Elective IV | 3 | |
| **Program Electives – Choose 2 (6 credits)** | | | |
| SCSR4453 | Network Security | 3 | |
| SCSR 4973 | Computer Network & Security Special Topics | 3 | |
| SECB3133 | Computational Biology I | 3 | |
| SECB3203 | Programming for Bioinformatics | 3 | |
| SECR5xx3/ SECP5xx3/ SECJ5xx3 | PRISMS Elective 1 | 3 | |
| SECR5xx3/ SECP5xx3/ SECJ5xx3 | PRISMS Elective 2 | 3 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **128** | |

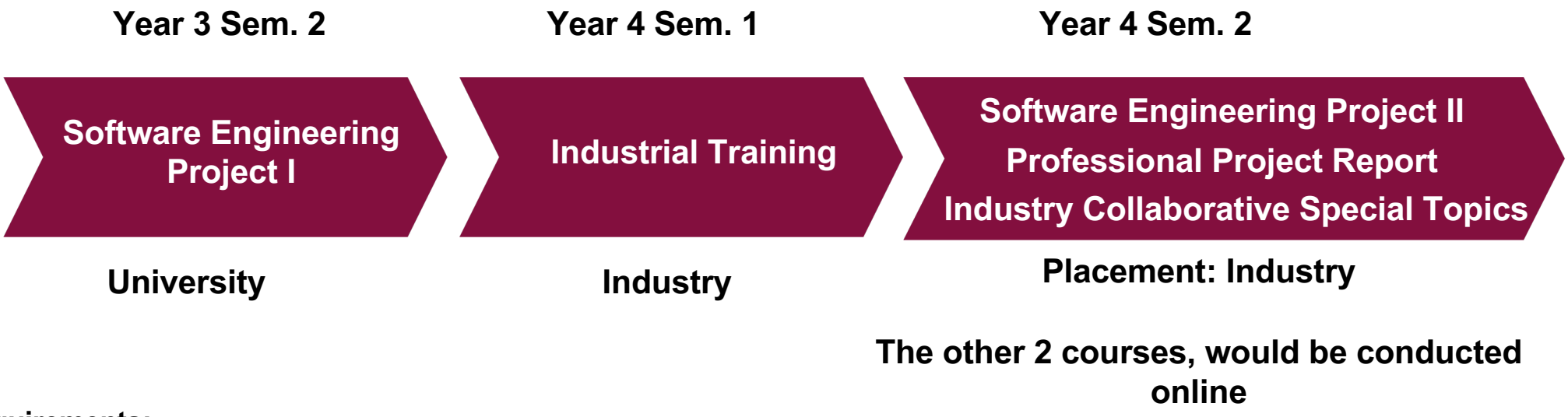
# PRISMS Elective

- PRISMS is a programme that integrates undergraduate high-level elective 5### courses with Master courses.

PRISMS - Program Integrasi Ijazah Sarjana Muda – Sarjana (Integrated Undergraduate - Postgraduate Program)

| YEAR 4: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE4214 | Software Engineering Project II | 4 | SCSE3242 |
| Sxxxxxx3 | Free Elective III | 3 | |
| Sxxxxxx3 | Free Elective IV | 3 | |
| **Program Electives – Choose 2 (6 credits)** | | | |
| SCSR4453 | Network Security | 3 | |
| SCSR 4973 | Computer Network & Security Special Topics | 3 | |
| SECB3133 | Computational Biology I | 3 | |
| SECB3203 | Programming for Bioinformatics | 3 | |
| SECR5xx3/ SECP5xx3/ SECJ5xx3 | PRISMS Elective 1 | 3 | |
| SECR5xx3/ SECP5xx3/ SECJ5xx3 | PRISMS Elective 2 | 3 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **128** | |

# Integrated Final Year Project - Internship Collaboration




**The other 2 courses, would be conducted online**

**Requirements:**

i) CGPA > 3.67 during Year 3 Sem 1
ii) Only 30 students will be selected.

# UTM Professional Skills Certificate




Must register and PASS all 3

Choose 1




Choose 1

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# 03 THINGS A STUDENT MUST KNOW

# Academic Advising

- All students are assigned an academic advisor for the following purposes:
    i. To assist students in understanding university policies, procedures and regulations.
    ii. To provide information on academic programmes, institutional support services and resources.
    iii. To assist students in planning, monitoring and evaluating their educational plan towards degree completion and development of decision-making skills.

- Students MUST meet with their academic advisor at least once each semester.

# OTHER SPECIAL PROGRAMMES FOR UG

- First Year Experience
- Cross Campus Programmes
- Global Outreach Programmes
- UTM Mobility (Outbound) – Best is during Yr 3 Sem 1
- Minor Programmes

# Global Outreach Programme






# MECHANISM of INFORMATION DISBURSEMENT

- Whatsapp or Telegram group
  - Year 1 group
  - Program group
  - Courses group
  - Academic Advisory group

Batch group

https://t.me/+I_5GupR2raI2MGZI

# NEW STUDENTS DETAILS



# ENTRY SURVEY



# NEW STUDENTS KIT



https://online.fliphtml5.com/nyrfd/kslq/?1726674883326

# Ms Rohani’s Telegram group






innovative ● entrepreneurial ● global

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# Good Luck

# Thank You

